南京工程学院
NANJING INSTITUTE OF TECHNOLOGY

OULUN YLIOPISTO

Programming #4, 2026

Exercise #1: Setup, Project requirements, and HTML

Set up your environment

You should've already installed the following extensions. If you haven't done so, then do it right away.

Install the following extensions:

- **Live Preview**: Provides a local side-by-side preview of your HTML document. Auto reloads as you code.
- **Intellisense for CSS class names in HTML**: Autocomplete CSS class names and ID suggestions from your stylesheets, so that you don't have to memorize them. Also works with Bootstrap framework.
- **Prettier**: Auto formats your JavaScript code for readability. Helps you develop good coding habits
- **Bootstrap 5 Quick snippet**s: Just like above, but for Bootstrap. Also offers other benefits. Make sure it's Bootstrap 5 and not Bootstrap 4, as we will be using Bootstrap 5 in this course.
- **Vue Official :** Similar to above, but for Vue. This extension is for Vue 3 only, as we will be using Vue version 3 in our course.

Website Planning

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Website Planning

It’s always a good idea to plan before implementing something. This applies to website/UI as well, especially when you are going to use so many different technologies.

For now, and in this exercise, you will first learn and understand a very general checklist. More specific checklist, associated with JavaScript, Bootstrap, and Vue will be dealt with later in the course, once you are familiar with those technologies.

For now, we’ll deal with HTML and CSS only.

Website Planning Checklist

Parameter	Description
Website purpose	Why are you making the website? What will the website help you, or your company, achieve?
Target audience	Whom are you targeting with your website? What type of users will be attracted to your website and engage with it? Describe the users based on their age, gender, demographics, etc.
Platform	Will your website be usable on computers (desktops and laptops) and mobile devices (cellphones, tablets, etc.)? If not, then why not?
Site map	How many pages will your website have? How are these webpages organized?
Wireframe	What features will be displayed on each webpage and how? It's about the layout of the different page elements
Graphics	What graphics will you use in your website? These are your images, animations, video, audio, etc.
Typography	What font styles will you use in your website?
Accessibility	What features will you add to make your website accessible to people with disabilities, e.g., people with sight or hearing issues

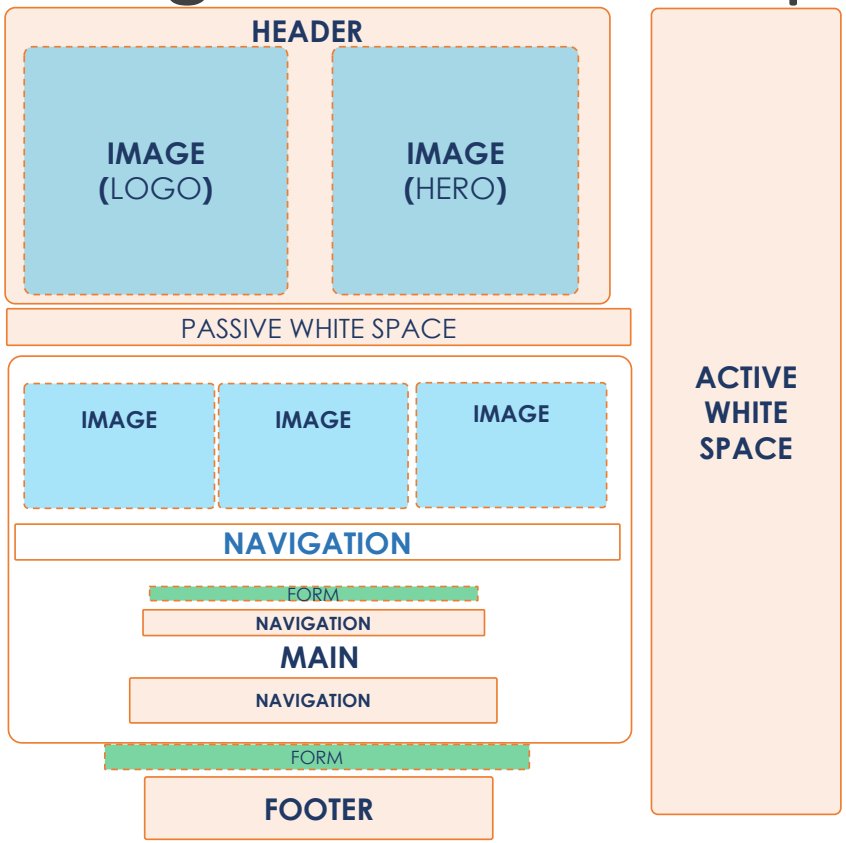
Source: Based on the Checklist available in the book **Responsive Web Design with HTML and CSS**, *9th Edition,* by **Jessica Minnick**

Website Planning Checklist Example #1

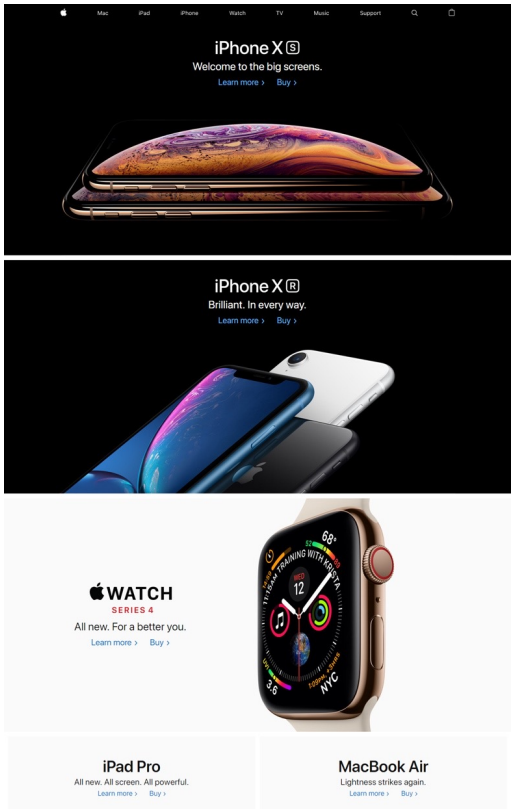
Website Planning Checklist Example #1 (ctd.)

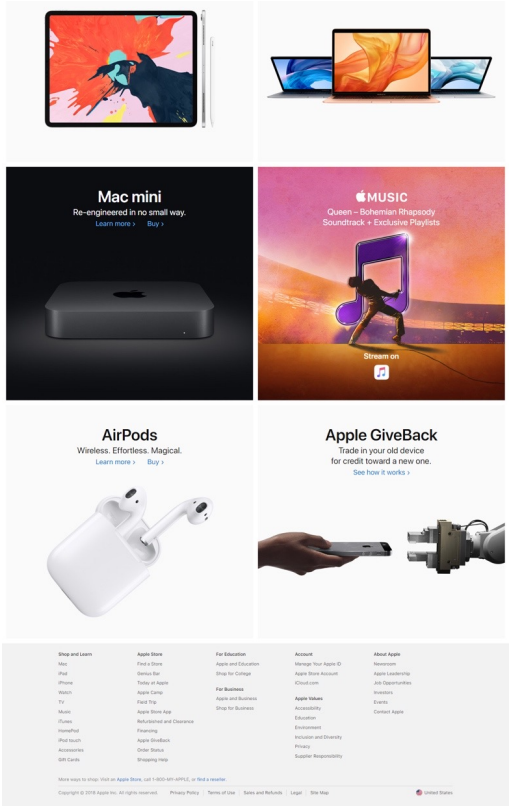
Parameter	Description
Website purpose	To promote Apple products, inform users of product features, and enable user navigation to different products and services available across the website, including purchasing and educational resources.
Target audience	Tech enthusiasts, students, professionals, current and potential Apple customers across the world.
Platform	Designed mainly for desktop browsers. No mobile devices supported.
Site map	Pages include product highlights, support, education, store, news, developer resources, and more. They are all organized via prominent link menus.
Wireframe	*Refer to the figure on the next slide.*
Graphics	Static images. No animations or multimedia, except simple GIFs.
Typography	Plain fonts. No custom typography or styling used
Accessibility	No accessibility features available.

Website Planning Checklist Example #1 (ctd.)



Website Planning Checklist Example #2

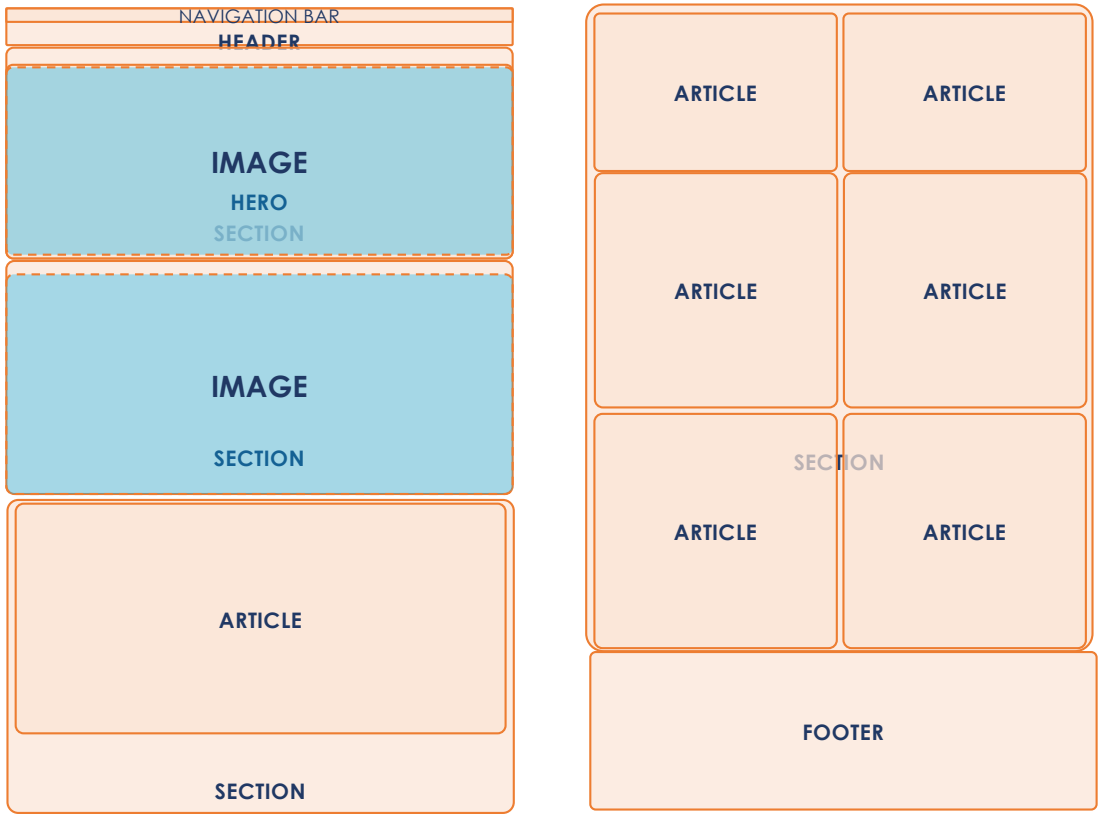




Website Planning Checklist Example #2 (ctd.)

Parameter	Description
Website purpose	To promote Apple flagship products, inform users of product features, and enable user navigation to different products and services available across the website, including purchasing and educational resources. Promote innovation.
Target audience	Tech enthusiasts, students, professionals, creatives, music lovers, current and potential Apple customers across the world.
Platform	Fully responsive for all devices. Mobile-first design enables different platform compatibility
Site map	Hierarchical. Top navigation bar. Each section links to dedicated product pages.
Wireframe	*Refer to the figure on the next slide.*
Graphics	Large high-resolution images. Minimalist background. Products front and center. Light effects, shadows, and clean visuals.
Typography	Apple's proprietary typeface (San Francisco) used.
Accessibility	Semantic HTML5 used. Scalable fonts and images. Includes ARIA roles and screen reader support.

Website Planning Checklist Example #2



Course Project

General requirements

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Course Project general requirements

You will be developing a **Single Page Application (SPA)** as part of your group project for this course. The general requirements for this project is as follows:

The **SPA** must use the following technologies:

- **Semantic HTML** for structure and accessibility
- **CSS** for styling
- **JavaScript** for interactivity
- **Bootstrap** for responsive layout and components
- **Vue.js** for reactivity and component-driven design

The **SPA** must:

- Be **responsive, interactive**, and **reactive**
- Follow the **MVC design pattern** (model = data, view = UI, controller = logic))
- Use **reusable** and **modular** elements / components
- Include a **form** to search/filter content on the site
- **(OPTIONAL)** Include other forms like **login** or **feedback**
- Perform **form validation** for all user inputs
- Respond to user actions by **processing data and updating the view** (e.g., using JavaScript or Vue methods)

Course Project general requirements (ctd.)

- Display a collection of **similarly themed content** (e.g., product list, events, items).
- Include a **hero** section, highlighting featured content or major announcement
- Implement at least one visual effect such as **carousal**, **parallax** section, or **animated** banner
- Contain **navigation** links, a **logo**, and **multimedia** (at least images)
- Use **buttons** for actions like submitting forms or triggering interactivity
- Include one clearly defined **main** section containing the SPA's primary content

The final project must be:

- **Tested and fully funcitonal**
- **Documented with a final project report**
- **Publicly accessible online (e.g., GitHub Pages)**

Specific functional and technical requirements for JavaScript, Bootstrap, and Vue will be shared as we progress in the course and as you become more familiar with these technologies

SPA references

Following are few sites that will give you some idea about a typical SPA:

- **Luckin Coffee**: https://www.luckincoffee.com/
  - The product banner is a **hero** section
  - Animations about the products are the **multimedia**
  - Drink listing with visuals are the similarly themed content, also known as **collections**
  - Menu for **navigation**
  - Works smoothly on cellphones, which means it's **responsive**
- **Apple's iPhone product page**: https://www.apple.com/iphone/
  - Has **hero** section, **responsive** design, uses **multimedia**, **interaction** by scrolling, **forms**, **styling** with **parallax** effect, **modularity**, and has **forms**
- **Starbucks Reserve**: https://www.starbucksreserve.com/
  - Has **hero** section, **responsive** design, uses **multimedia**, **navigation** and **buttons**, **forms**, **styling** with **parallax** effect, **modularity**, and has **forms,** and uses some frameworks like Bootstrap

2025 Project references

Following are some of the projects submitted last year:

- **Visually best looking**: https://peirangu.github.io/P4-Project-MovieDB/
- **Some nice use of UI conepts they learned**: https://aria1211.github.io/OPH_Team_Project/
- **Creative and out-of-the-box use of UI concepts they learned**: https://thesolitaryshadow.github.io/GG_Bond_group_work/
- Nice all-rounder: https://panyiifan.github.io/Project/

2025 Project Topics

Following are the projects topics from last year:

- Travel booking
- Movie review
- Restaurant booking
- Conan anime
- A Robotics company
- Green company
- Car sales platform
- Online Library
- Fitness
- Hospital appointment system
- Camera sales platform

YOU CANNOT USE ANY OF THESE TOPICS

2026 Project Topic

- Finalize your group soon and register it in Moodle
- Discuss a suitable project topic in your group
- Have ONE MEMBER from your group register the topic in Moodle in **Project Topic Registration**

Deadline for Topic Registration: April 8, Midnight

Exercise #1

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Shinchan site design

- Use the ”**Shinchan – Homepage**” and ”**Shinchan site video**” files to understand the site design
- Using only **HTML5**, design the site
  - All the assets needed for the site is available in the **assets** folder (in **assets.zip**) shared on Moodle
- You must use HTML5 semantic tags
  - You may also need **` `**. It’s a **non-breaking space** used to provide space between elements, but doesn’t break the line. It’s like using a **Spacebar** on your keyboard.
- Name your root folder ”**EX2-HTML**”
  - Make sure all the files and folder are inside this root folder, including the **`index.html`** file

Some concepts you may need: `colspan`

- Used in tables to combine multiple columns into one. For example:

colspan = “4”			
1	2	3	4

```
<table width="100%" border="1">
  <thead>
    <tr>
      <th colspan="4">colspan = "4"</th>
    </tr>
    <tr>
      <td>1</td>
      <td>2</td>
      <td>3</td>
      <td>4</td>
    </tr>
  …
```

colspan = “3”		
1	2	3

```
<table width="100%" border="1">
  <thead>
    <tr>
      <th colspan="3">colspan = "3"</th>
    </tr>
    <tr>
      <td>1</td>
      <td>2</td>
      <td>3</td>
    </tr>
  …
```

`rowspan` works the same way for rows

Some concepts you may need: `radio`

Used in forms for radio button inputs. For example:

Your Gender: ○ Male ○ Female

```
<label>Your Gender: </label>
<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label>
```

- `input type = "radio"`: Indicates that it's a radio button
- `id = "male"`: Provides unique ID to the input. This ID is also in the label. So, clicking the label will select the radio button
- `name = "gender":` Makes it part of the group "gender". This ensures that only one button can be picked in this group
- `value = "male"` : When the form will be submitted, and if "Male" was picked, then the form will send this value to the server, or any backend logic.
- `for= "male":` Is how the label connects to the input with the ID "male"

Submission Instructions

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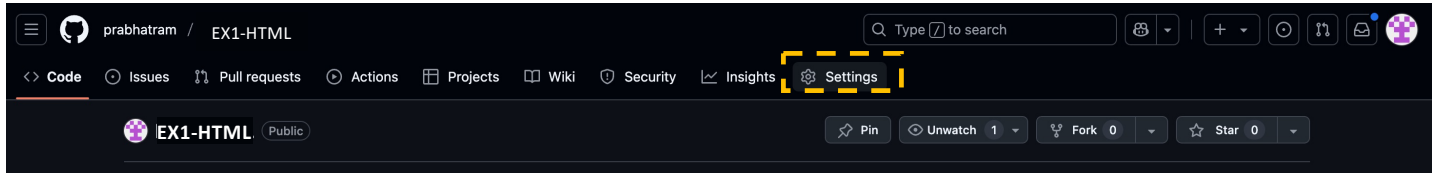
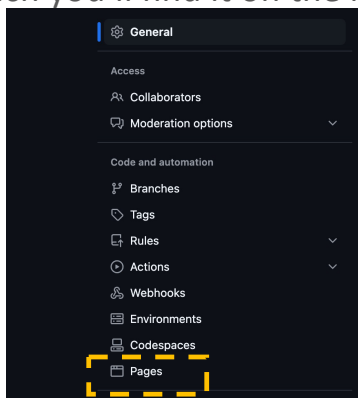
Submission

- Commit and push your exercise to your GitHub account
- Go to the repository that is created in your account. Go to **Settings**

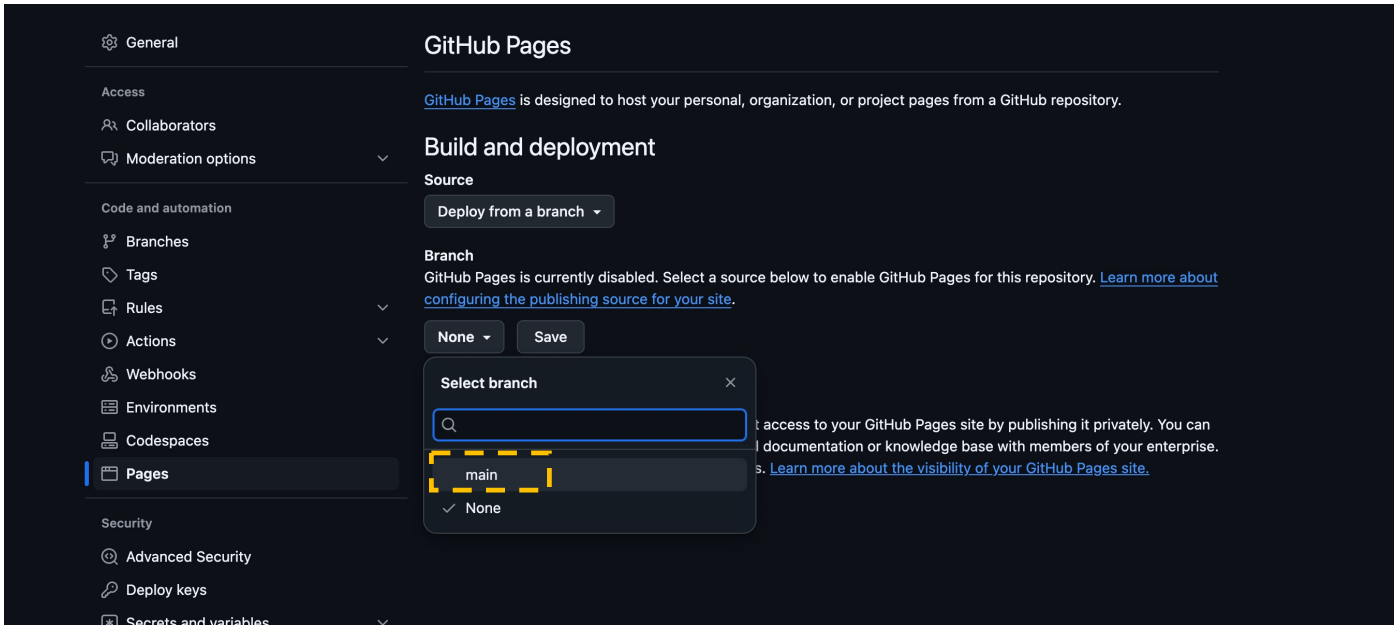


- In the next page, go to **Pages,** which you'll find it on the left side of the page.



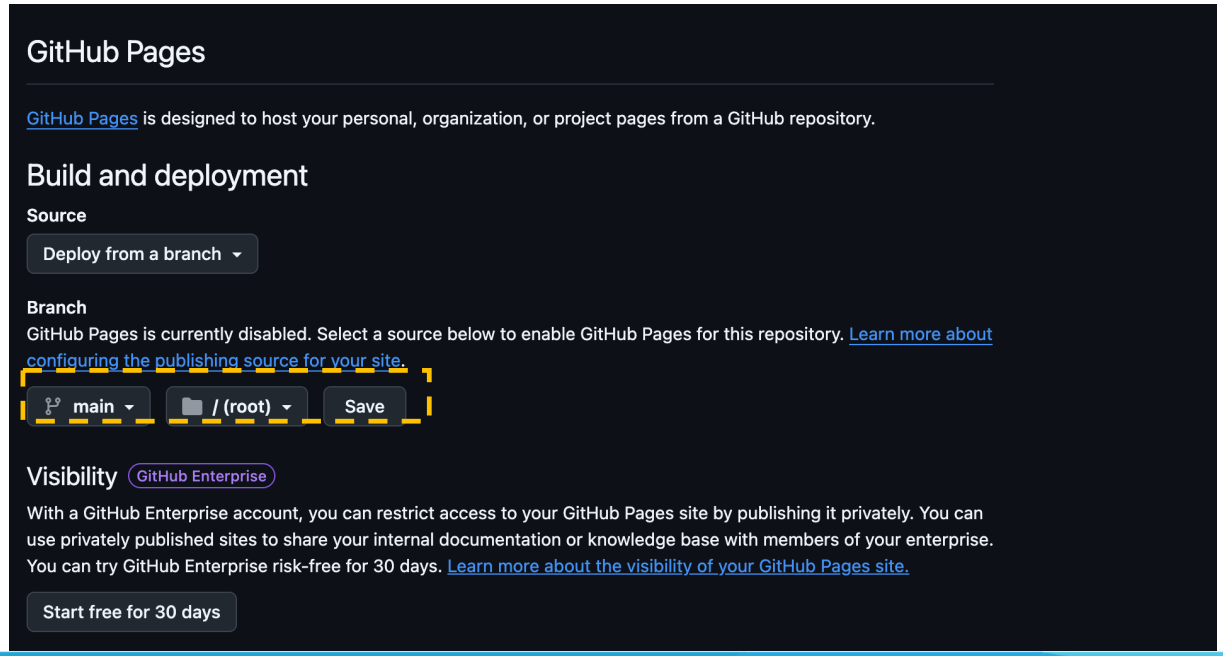
Submission (ctd.)

- **GitHub Pages** will open on the right. There, look for the **Branch** section. Click on the dropdown menu and select `main` branch



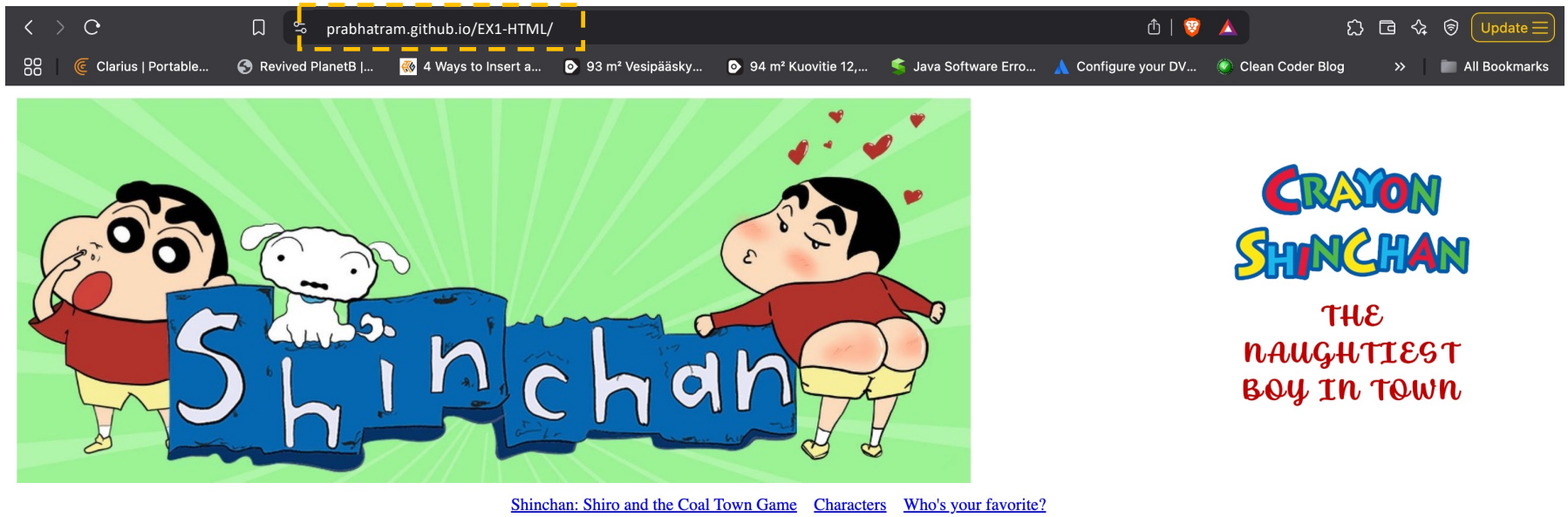
Submission (ctd.)

- After selecting the **main** branch, the section should look like what you see below in the screenshot. This makes sure GitHub pages is able to find your `index.html` file when it renders it. Click **Save**



Submission (ctd.)

- Go in your browser and type **`username.github.io/EX1-HTML/`** and hit Enter. Remember to replace **`username`** with your own GitHub account username. The site you've designed will be displayed. It might take some time the first time you open the page, as GitHub is trying to render everything



Submission (ctd.)

- In **Moodle's Exercise 1 – HTML** submission box, just submit the link **`username.github.io/EX1-HTML/`**.
- Remember to replace **`username`** with your own **GitHub account username.**

That’s all for this exercise